

Chapter 4

Lecture 4: Homeomorphisms and connectedness

Definition 4.0.1. A continuous bijective map $f : X \rightarrow Y$ is called a homeomorphism if f^{-1} is also continuous.

Remark "homeomorphism" is not the same as "homomorphism".

Definition 4.0.2. $X \simeq Y$ are homeomorphic or topological equivalent if there exists a homeomorphism $f : X \rightarrow Y$

Definition 4.0.3. Two metric spaces (X, d_X) and (Y, d_Y) are called isometric if there exists a bijective map $f : X \rightarrow Y$ such that for all $x_1, x_2 \in X$, $d_Y(f(x_1), f(x_2)) = d_X(x_1, x_2)$. (f is called an isometry)

Exercise Prove that every isometry is a homeomorphism, but some homeomorphisms are not isometries.

Example 4.0.1. Let $D = \{(x, y) \in \mathbb{R}^2 | x^2 + y^2 \leq 1\}$ be the unit disk in $\mathbb{R}^2$ with the Euclidean metric. Let $S^1 = \{(x, y) \in \mathbb{R}^2 | \max |x|, |y| = 1\}$ be the unit square in $\mathbb{R}^2$ with the Euclidean metric. Then D and S^1 are homeomorphic but not isometric.

Proof

The idea of prove homeomorphism is to map every radius of the disk to the corresponding line segment of the square.

Assume there exists an isometry $f : D \rightarrow S^1$. Note that the diameter of D is 2, while the diameter of S^1 is $2\sqrt{2}$. This contradicts the definition of isometry. Thus, no such isometry exists.

Example 4.0.2. Another example of homeomorphism is the linear map from the interval $(0, 1)$ to (a, b) (by stretching and compressing).

Example 4.0.3. $(0, 1) \simeq (-\frac{\pi}{2}, \frac{\pi}{2}) \simeq \mathbb{R}$ by the tangent function.

The above example can be described graphically by a bowl (a circle centered at $(0, \frac{1}{2})$ with radius $\frac{1}{2}$ without the upper half) with radius $\frac{1}{2}$ on the real line.

Property 4.0.1. If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are continuous, then $g \circ f : X \rightarrow Z$ is continuous.

Corollary 4.0.1. If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are homeomorphisms, then $g \circ f : X \rightarrow Z$ is a homeomorphism.

Property 4.0.2. Homeomorphism $\simeq$ is an equivalence relation.

Definition 4.0.4. X is said to be path-connected if any two points $x, y \in X$ can be joined by a path: there exists a continuous map $f : [0, 1] \rightarrow X$ such that $f(0) = x$ and $f(1) = y$.

Definition 4.0.5. Let X be a metric space. X is said to be connected if one of the following equivalent conditions holds:

1. X cannot be represented as $X = U_1 \sqcup U_2$ where U_1, U_2 are non-empty open subsets of X .
2. X cannot be represented as $X = V_1 \sqcup V_2$ where V_1, V_2 are non-empty closed subsets of X .
3. There is no proper non-empty subset $U \subseteq X$ which is both open and closed in X .

Property 4.0.3. If X is path-connected and $f : X \rightarrow Y$ is a homeomorphism, then Y is also path-connected.

Proof Let $y_1, y_2 \in Y$. Since f is bijective, there exist $x_1, x_2 \in X$ such that $f(x_1) = y_1$ and $f(x_2) = y_2$. Since X is path-connected, there exists a continuous map $g : [0, 1] \rightarrow X$ such that $g(0) = x_1$ and $g(1) = x_2$. Consider the map $h = f \circ g : [0, 1] \rightarrow Y$. Since both f and g are continuous, h is continuous. Moreover, $h(0) = f(g(0)) = f(x_1) = y_1$ and $h(1) = f(g(1)) = f(x_2) = y_2$. Thus, there exists a continuous path in Y connecting y_1 and y_2 , proving that Y is path-connected. ■

Theorem 4.0.1 (Intermediate Value Theorem). *Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. For any c between $f(a)$ and $f(b)$, there exists some $x \in [a, b]$ such that $f(x) = c$.*

Example 4.0.4. $\mathbb{R} \setminus \{0\}$ is not path-connected.

Proof By IVT.

Theorem 4.0.2 (Intermediate Value Theorem for Path-Connectedness). *Let X be a path-connected space and $f : X \rightarrow \mathbb{R}$ be continuous. Suppose there exist $x_1, x_2 \in X$ such that $f(x_1) = c_1$ and $f(x_2) = c_2$. Then for any c between c_1 and c_2 , there exists some $x \in X$ such that $f(x) = c$.*

Proof Since X is path-connected, there exists a continuous map $g : [0, 1] \rightarrow X$ such that $g(0) = x_1$ and $g(1) = x_2$. Consider the map $h = f \circ g : [0, 1] \rightarrow \mathbb{R}$. Since both f and g are continuous, h is continuous. Moreover, $h(0) = f(g(0)) = f(x_1) = c_1$ and $h(1) = f(g(1)) = f(x_2) = c_2$. By the Intermediate Value Theorem, for any c between c_1 and c_2 , there exists some $t \in [0, 1]$ such that $h(t) = c$. Let $x = g(t)$. Then $f(x) = f(g(t)) = h(t) = c$. Thus, there exists some $x \in X$ such that $f(x) = c$. ■

Property 4.0.4. *Let X be a metric space. Then IVT for X holds if and only if there is no continuous and surjective map $f : X \rightarrow \{0, 1\}$.*

Proof

We assume that X contains more than one point.

One direction is trivial.

Let's assume that the IVT doesn't hold for X .

Since the IVT does not hold for X , there exists $y_1, y_2 \in \mathbb{R}$ such that there exists y_3 between y_1 and y_2 which is not in the range of f . Then $f : X \rightarrow \mathbb{R} \setminus \{y_3\}$. Now we can easily define a continuous and surjective map $\tilde{f} : X \rightarrow \{0, 1\}$. ■

Property 4.0.5. *Let X be a metric space. X is connected if and only if IVT for X holds.*

Proof

The statement is equivalent to: X is connected if and only if there is no continuous and surjective map $f : X \rightarrow \{0, 1\}$.

Suppose there is a continuous and surjective map $f : X \rightarrow \{0, 1\}$. Then $f^{-1}(\{0\})$ and $f^{-1}(\{1\})$ are non-empty, disjoint, open subsets of X whose union is X . Thus, X is not connected.

Suppose that there is no continuous and surjective map $f : X \rightarrow \{0, 1\}$. If X is not connected, then there exist non-empty, disjoint, open subsets U_1, U_2 of X such that $X = U_1 \sqcup U_2$. We can define a map $f : X \rightarrow \{0, 1\}$ by setting $f(x) = 0$ if $x \in U_1$ and $f(x) = 1$ if $x \in U_2$. This map is continuous and surjective, contradicting our assumption. Therefore, X must be connected. ■

Corollary 4.0.2. *Let X be a metric space. If X is path-connected, then X is connected.*

Proof Since X is path-connected, IVT for X holds. Thus, X is connected. ■

Remark IVT holds for X doesn't implies X is path-connected.

Example 4.0.5. *This is an example of a connected space but not path-connected:*

$$X = \{(x, \sin \frac{1}{x}) | x > 0\} \cup (\{0\} \times [-1, 1]) \quad (4.1)$$